Prompt

Sampling

T2I Model

Policy Optimization

RL Algorithm

Samples & Prompt

Bottleneck-aware Constrained Reinforcement Learning for Multi-text Insulation

Step 1: Target Extraction & Instance Preprocessing

Prompt ⟶ $V_T = \{v_{t1}, v_{t2}, \cdots, v_{tn}\}$ ⟶ Normalization

Sample ⟶ OCR Model ⟶ $\mathcal{O}$ ⟶ Normalization

Step 2: Isolated Fuzzy Matching

$\cdots \mathcal{O}$

Sliding window: w

$$s_i = \max_{w \subseteq \mathcal{O}, |w| = |v_{ti}|} \left(1 - \frac{dist(v_{ti}, w)}{|v_{ti}|}\right)$$

Step 3: Insulation-aware Aggregation

$$R_{base} = (1 - \lambda_{bal}) \cdot \frac{1}{n} \sum_{i=1}^{n} s_i + \lambda_{bal} \cdot \min(s_1, \ldots, s_n)$$

Step 4: Anti-interference Penalty

Ratio: $\frac{l_{pred}}{l_{target}} \leq \delta$ — Yes: $\lambda_{noise} = 1.0$; No: $\lambda_{noise} = \frac{1}{\text{Ratio} - \delta + 1}$ ⟶ $R_{ocr} = \lambda_{noise} \cdot R_{base}$

Prompt: A cozy coffee shop interior with a chalkboard reading 'Brew', a mug with 'Love' written on it, a bag of beans labeled 'Fresh', and a poster on the wall saying 'Relax'.

Joint Attention Layer in MMDiT Blocks

Key

Query

Brew Love Fresh Relax

Text-to-Text Attention

Text-to-Image Attention

Quotation-guided Attention Gate

Image-to-Image Attention

Image-to-Text Attention

Attention Modulation

Noised Latent

Denoised Latent

× N

Decoder

